ABDULLAH YOUSAF KHAN

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EDUCATION

George Mason University — M.S. Data Analytics Engineering | GPA: 3.91 / 4.0 2026

Coursework: Data Management & Mining, Big Data Analytics, Bayesian AI, NLP with Deep Learning, Learning from Data, Analytics & Modeling

Beaconhouse National University — B.Sc. Computer Science 2023

Certifications: HIPAA Certified (2023) • IBM Generative AI Specialist (2024) • Google Data Analytics Professional Certificate (2024)

TECHNICAL SKILLS

Languages: Python, SQL, R, C++

AI / ML: GenAI & LLMs (GPT, Claude, Llama 3, Mistral), RAG (LangChain, LlamaIndex), Vector DBs (FAISS), Fine-tuning, Prompt Engineering, LangSmith, NLP, Scikit-learn, XGBoost, TensorFlow

Data & Cloud: Python, SQL, R, PySpark, Pandas, NumPy, SciPy, Matplotlib, AWS (SageMaker), Databricks, Hadoop, Tableau, Power BI, Excel (Adv), Git

Delivery & Domains: Agile/Scrum, Jira, YouTrack, BRD/PRD Authoring, System Design, Healthcare (HIPAA, HL7/FHIR, Insurance), Marketplaces, A/B Testing, Risk/Fraud Mitigation

EXPERIENCE

Scrum Master & Data Analyst — Global Terrorism Trends & Analysis Center / George Mason University 2025 — 2026

Fairfax, VA | Capstone Engagement

- Led 5-person team as **Scrum Master** delivering data products for the Global Terrorism Trends & Analysis Center; ran sprints, retros, and stakeholder demos in YouTrack across a 5-month engagement.
- Performed temporal trend analysis on **62,000+ global terrorism incidents** in Python (Pandas, GeoPandas, Folium); surfaced insights adopted into global terrorism policy briefings.
- Led research and scoped analytical deliverablesd aligned with GTTAC's congressional reporting mandate; coordinated weekly with project POC Dr. Mahmut Cengiz (GMU Schar School of Policy and Government).

Product Executive — Dubizzle Labs 2024

Onsite | Marketplace serving 3M+ Users

- Shipped **5+ product features end-to-end**; authored PRDs and managed sprint execution in Jira, improving development turnaround by **22%**.
- Built KPI dashboards tracking DAU, retention, and session metrics; analyzed user behavior to guide feature prioritization across 3 launches.
- Conducted competitive pricing and market analysis across 10+ platforms; optimized product support workflows with engineering, cutting issue-resolution time by **30%**.

Business Analyst — Alivia Analytics 2023 — 2024

Onsite | US Healthcare Insurance Analytics

- Led end-to-end onboarding for one of the **largest US healthcare insurance agency in firm history**; authored BRDs and User Specification documents under HIPAA-compliant workflows.
- Helped identify fraud patterns using insurance claims data, contributing to **$2.7M in company savings** on insurance fraud prevention.
- Analyzed business workflows and optimized client transitions, reducing onboarding friction by **25%**; designed user-centered product features based on operational needs.

PROJECTS

Medical Fact-Checker RAG System Python • LangChain • FAISS • Ollama • Mistral

- Built end-to-end **Retrieval-Augmented Generation (RAG) pipeline** using LangChain, FAISS (768-dim embeddings), and a locally-hosted Mistral LLM — grounding clinical responses in verified WHO source data and eliminating hallucinations.
- Engineered hub-scraping (BeautifulSoup4), context-aware recursive chunking, and prompt engineering frameworks to optimize retrieval accuracy; results exported to structured CSV.

Hospital Performance & Patient Flow Analytics Dashboard

- Queried and analyzed hospital operations data using **SQL** to evaluate bed occupancy, patient wait times, and resource utilization.
- Built a **Tableau dashboard for healthcare KPIs** enabling visibility into patient flow and bottlenecks.
- Applied **time-series forecasting** with **89% prediction accuracy** for patient admissions.
- Identified inefficiencies in resource allocation and proposed data-driven operational improvements.
- Showcased how analytics improves hospital performance, patient experience, and capacity planning.

Public Health Forecasting & Epidemiological Analytics (COVID-19)

- Analysed socio-economic and healthcare infrastructure data using **Python and SQL** to identify factors influencing COVID-19 spread across districts.
- Engineered features from literacy rates, population density, healthcare access, and facility counts to model regional vulnerability.
- Built a time-series **forecasting** model to predict the third wave trajectory and growth patterns.
- Performed **exploratory data analysis to uncover correlations** between infrastructure gaps and infection rates.
- Demonstrated how **data analytics informs public health planning**, resource allocation, and outbreak preparedness.

Customer Segmentation, Churn Prediction & Campaign Analytics

- Performed **RFM analysis** and **k-means clustering** on **50K+ customers** to identify 6 actionable segments.
- Built a **Random Forest churn model** with **86% accuracy** to predict retention risk.
- Designed a **Tableau dashboard** tracking **campaign ROI improvement** and conversion funnel metrics.
- Translated insights into data-driven marketing and retention strategies.

Public Sentiment Intelligence using NLP

- Collected and preprocessed **social media text data** in Roman Urdu using **Python and NLP techniques** including tokenization, normalization, and stop-word removal.
- Engineered linguistic features and built a **sentiment classification model** to categorize opinions as positive, negative, or neutral.
- Addressed challenges of **non-standardized language and transliteration** through custom preprocessing pipelines.
- Analyzed sentiment trends to extract **actionable insights on public opinion and user perception**.
- Demonstrated how NLP can be applied for **customer insight, brand monitoring, and opinion mining** in low-resource languages.

LLM Meta-Evaluation Harness *(Python, Streamlit; OpenAI/Anthropic/Ollama)*

- Built a config-driven evaluation pipeline that scores LLM-generated long-form QA answers with lexical (ROUGE/METEOR), semantic (BERTScore), and LLM-as-judge metrics, then benchmarks each against ~300 human-labeled gold answers to measure which methods actually predict human judgment.
- Found LLM-as-judge scores correlate with human ratings far more strongly than traditional metrics (Spearman 0.76–0.86 vs. 0.27–0.32 for ROUGE/METEOR/BERTScore), quantifying why cheap automated metrics are unreliable for long-form generation eval.
- Audited LLM judges for position, verbosity, and self-preference bias using metrics from Zheng et al. (2023); measured 7.7–15.3% pairwise verdict-reversal rates under answer-order swap and applied swap-augmented calibration (Wang et al., 2023) to eliminate it, reporting the 12–16% tie-rate cost of doing so.
- Reproduced a published LLM-judge position-consistency finding and extended the analysis to a novel long-form QA domain.

Construction Project Cost, Timeline & Resource Analytics

- Analyzed construction project data using **SQL and Python** to monitor **budget vs actual cost** and **schedule variance**.
- Built a **Tableau dashboard** highlighting **schedule variance and cost overruns**.
- Identified delay patterns and cost drivers through exploratory data analysis.
- Provided insights to support **project planning, cost control, and operational efficiency**.
- Illustrated how analytics enhances **construction project management and decision-making**.

Real Estate Market Valuation & Investment Analytics

- Analysed a **Lahore real estate dataset** using **Python, Pandas, and SQL** to evaluate factors influencing property prices.
- Performed **feature engineering and correlation analysis** to identify key drivers such as location, area, amenities, and property type.
- Built a **regression model** predicting property value with **87% R^2 score**.
- Conducted exploratory data analysis to uncover **pricing trends, demand patterns, and neighbourhood valuation differences**.
- Demonstrated how predictive analytics supports **real estate investment decisions, pricing strategy, and market assessment**.

Credit Card Fraud Detection & Risk Analytics Dashboard

- Analyzed **280K+ transactions** using **SQL & Python** to identify anomalous spending patterns and fraud indicators.
- Built a **logistic regression model**with **86% accuracy** and engineered risk features achieving high recall for suspicious activity.
- Developed a **Power BI dashboard** tracking fraud rate, risk scores, geographic hotspots, and transaction trends.
- Defined audit-focused **risk KPIs** to support fraud prevention and financial risk assessment.